

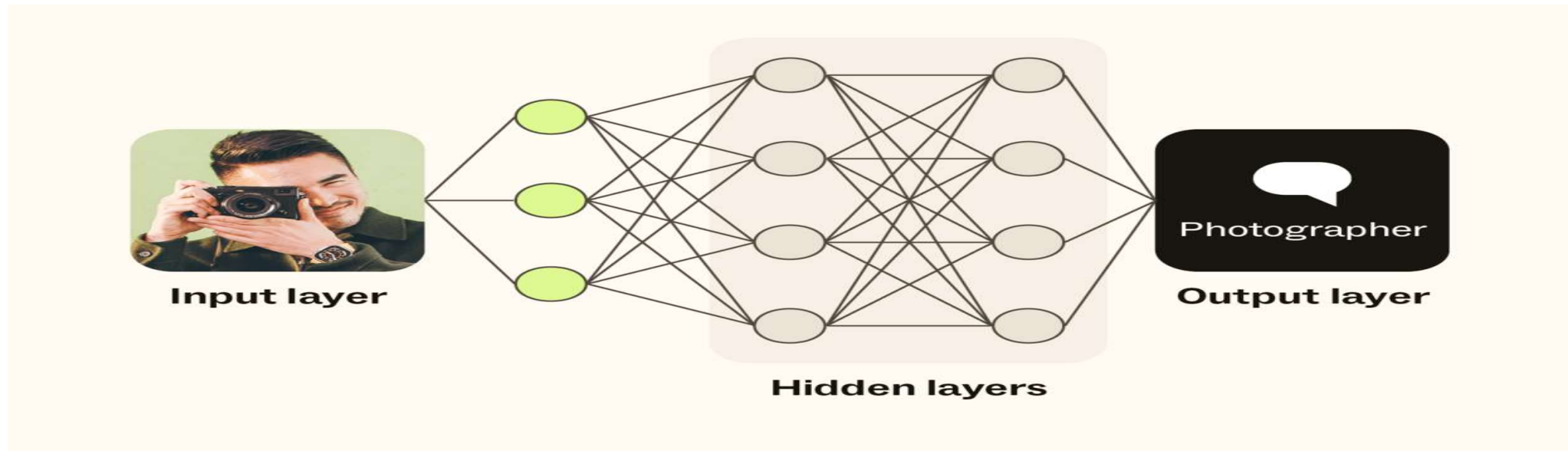


# INTRODUCTION TO DEEP LEARNING

Deep Learning

# What is Deep Learning?

- The branch of machine learning that is based on artificial neural network architecture.
- An artificial neural network or ANN uses layers of interconnected nodes called neurons that work together to process and learn from the input data.



# Why Deep Learning?

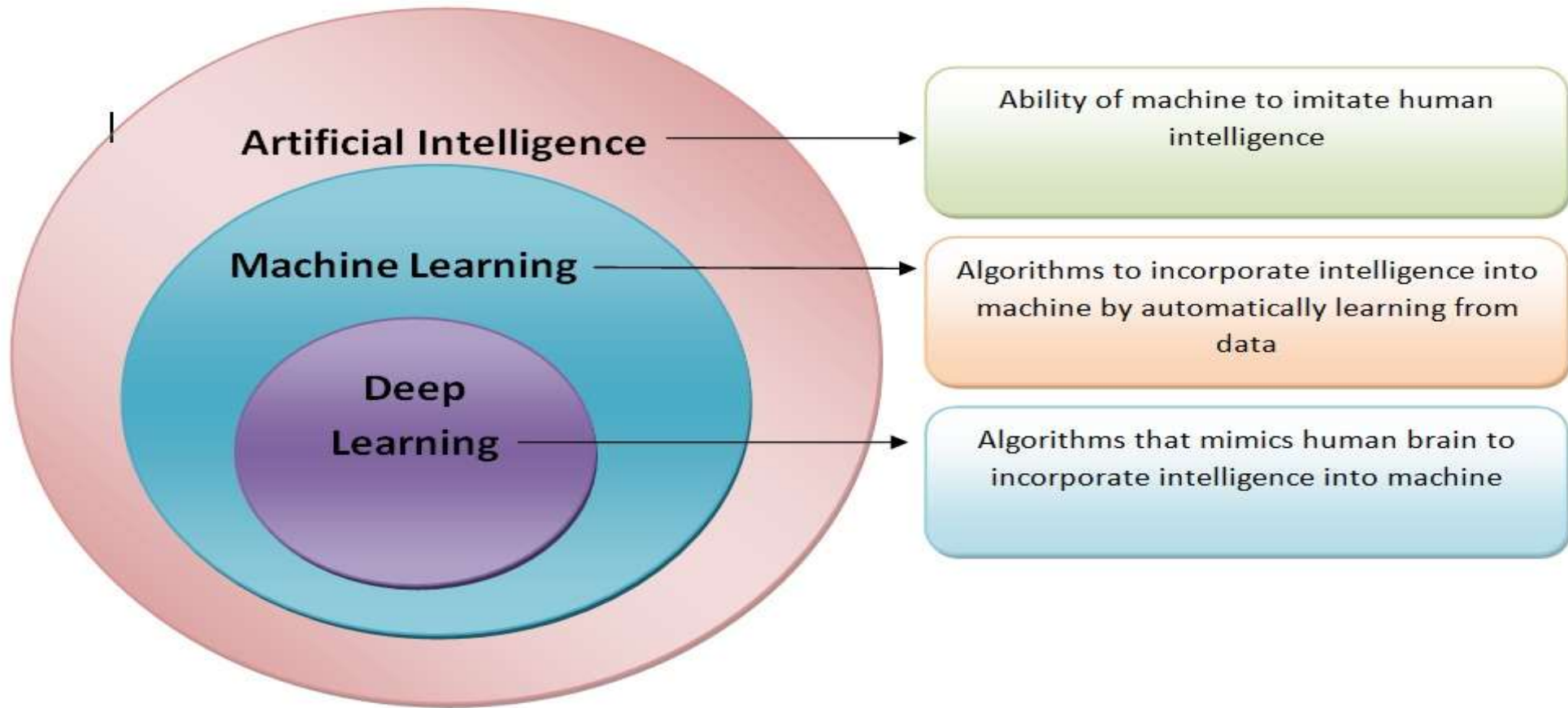
## **Limitations of traditional machine learning algorithms**

- not good at handling high dimensional data.
- difficult to do feature extraction and object recognition.

## **Advantages of deep learning**

- DL is computationally expensive, but it is capable of handling high dimensional data.
- feature extraction is done automatically.

# Comparison of AI with other Technologies



# Comparison of AI with other Technologies

| Artificial Intelligence                                                 | Machine Learning                                                             | Deep Learning                                                                                  |
|-------------------------------------------------------------------------|------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| AI simulates human intelligence to perform tasks and make decisions.    | ML is a subset of AI that uses algorithms to learn patterns from data.       | DL is a subset of ML that employs artificial neural networks for complex tasks.                |
| AI may or may not require large datasets; it can use predefined rules.  | ML heavily relies on labeled data for training and making predictions.       | DL requires extensive labeled data and performs exceptionally with big datasets.               |
| AI can be rule-based, requiring human programming and intervention.     | ML automates learning from data and requires less manual intervention.       | DL automates feature extraction, reducing the need for manual engineering.                     |
| AI can handle various tasks, from simple to complex, across domains.    | ML specializes in data-driven tasks like classification, regression, etc.    | DL excels at complex tasks like image recognition, natural language processing, and more.      |
| AI algorithms can be simple or complex, depending on the application.   | ML employs various algorithms like decision trees, SVM, and random forests.  | DL relies on deep neural networks, which can have numerous hidden layers for complex learning. |
| AI may require less training time and resources for rule-based systems. | ML training time varies with the algorithm complexity and dataset size.      | DL training demands substantial computational resources and time for deep networks.            |
| AI systems may offer interpretable results based on human rules.        | ML models can be interpretable or less interpretable based on the algorithm. | DL models are often considered less interpretable due to complex network architectures.        |
| AI is used in virtual assistants, recommendation systems, and more.     | ML is applied in image recognition, spam filtering, and other data tasks.    | DL is utilized in autonomous vehicles, speech recognition, and advanced AI applications.       |

# Application of Deep Learning

- Computer Vision

**Classification**



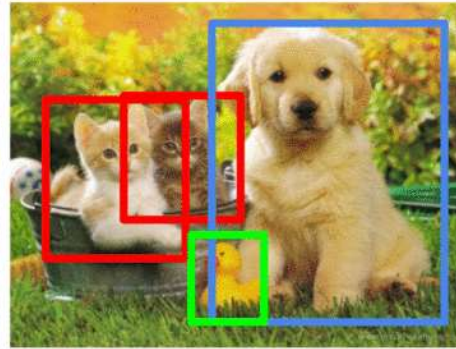
CAT

**Classification  
+ Localization**



CAT

**Object Detection**



CAT, DOG, DUCK

**Instance  
Segmentation**



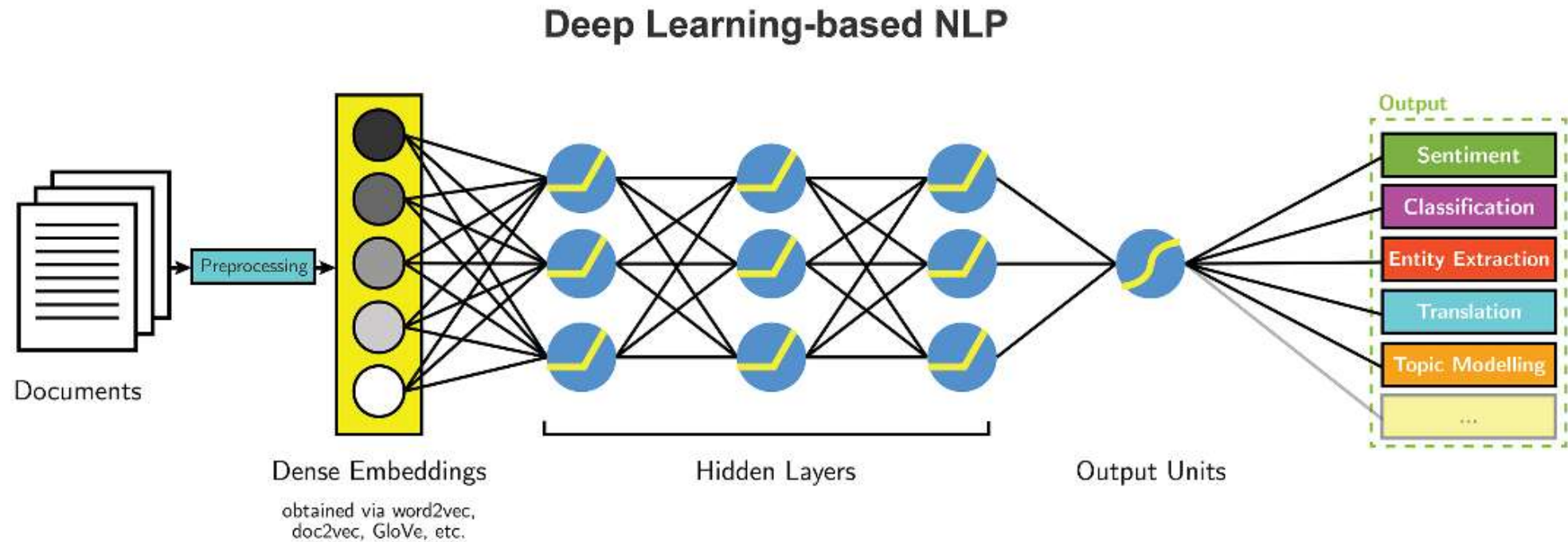
CAT, DOG, DUCK

Single object

Multiple objects

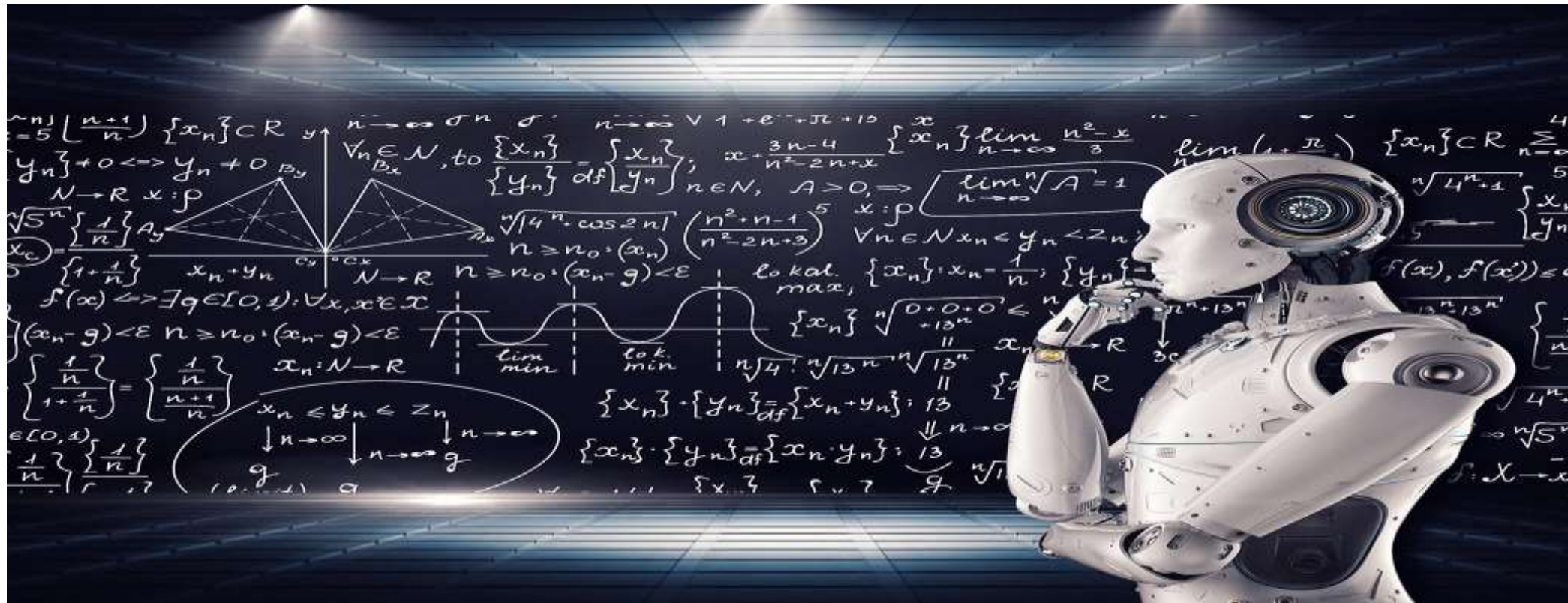
# Application of Deep Learning

- NLP(Natural Language Processing)



# Application of Deep Learning

- Robotics



# Application of Deep Learning

- Gaming



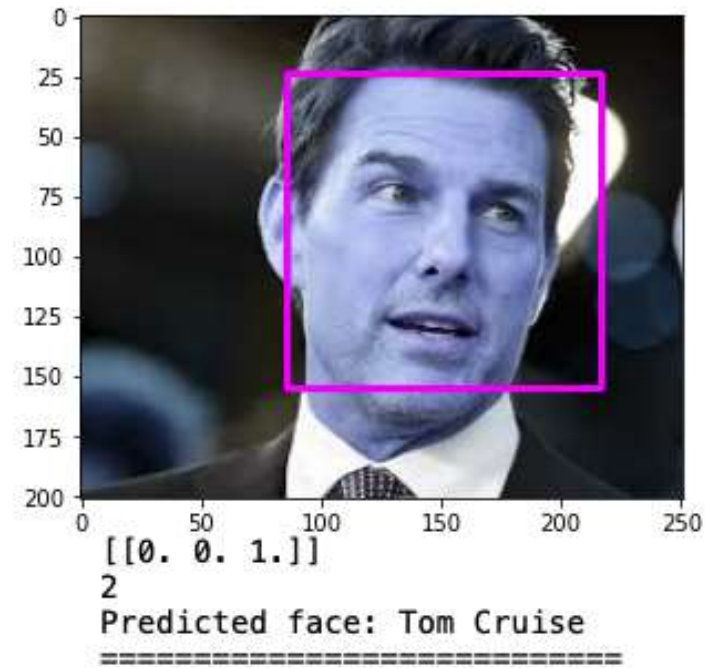
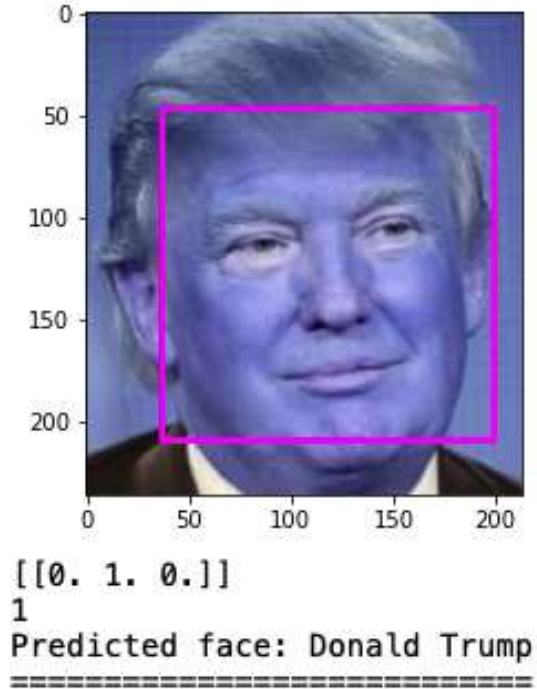
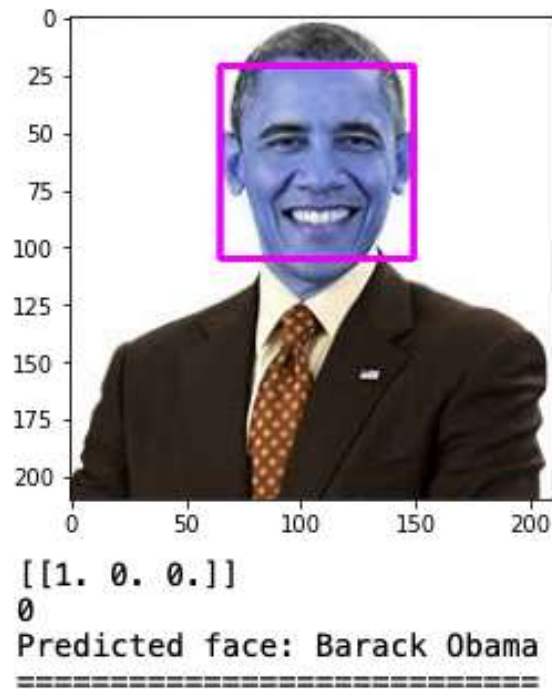
# Application of Deep Learning

- Healthcare



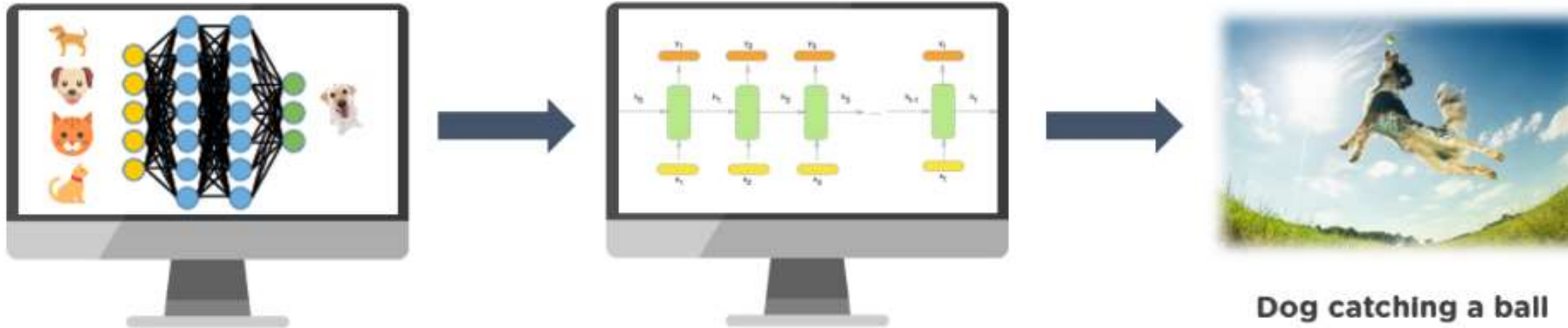
# Application of Deep Learning

- Face Recognition



# Application of Deep Learning

- Image Captioning



# Evolution of Neural Networks

- Since the 1940s, there have been a number of noteworthy advancements in the field of neural networks:

## **1940s-1950s: Early Concepts**

- Neural networks began with the introduction of the first mathematical model of artificial neurons by McCulloch and Pitts. But computational constraints made progress difficult.

## **1960s-1970s: Perceptrons**

- This era is defined by the work of Rosenblatt on perceptrons. Perceptrons are single-layer networks whose applicability was limited to issues that could be solved linearly separately.

# Evolution of Neural Networks

## **1980s: Backpropagation and Connectionism**

- Multi-layer network training was made possible by Rumelhart, Hinton, and Williams' invention of the backpropagation method. With its emphasis on learning through interconnected nodes, connectionism gained appeal.

## **1990s: Boom and Winter**

- With applications in image identification, finance, and other fields, neural networks saw a boom. Neural network research did, however, experience a “winter” due to exorbitant computational costs and inflated expectations.

# Evolution of Neural Networks

## **2000s: Resurgence and Deep Learning**

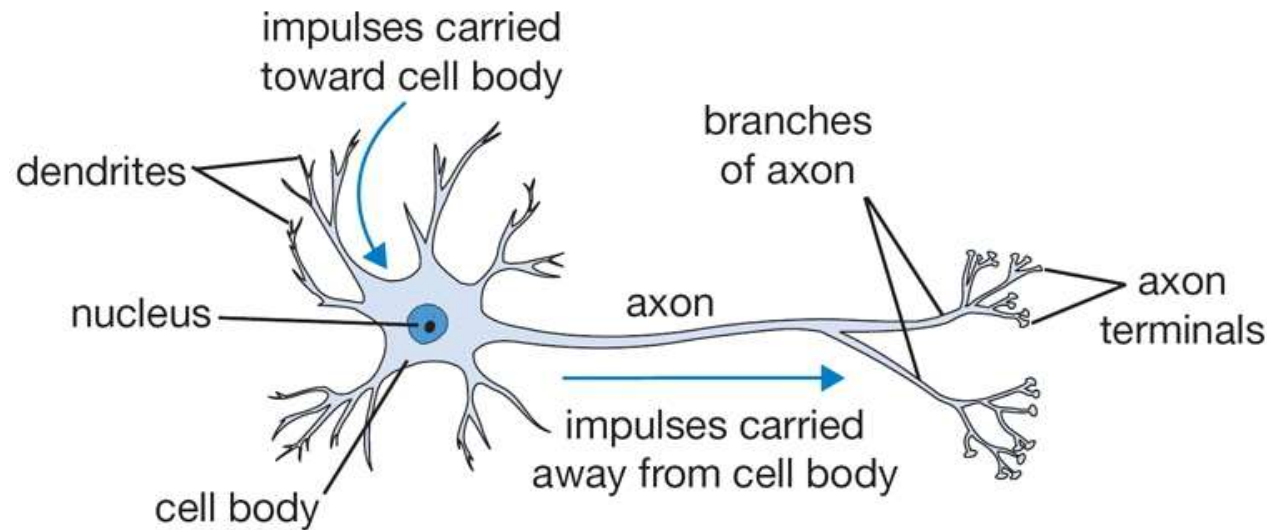
- Larger datasets, innovative structures, and enhanced processing capability spurred a comeback. Deep learning has shown amazing effectiveness in a number of disciplines by utilizing numerous layers.

## **2010s-Present: Deep Learning Dominance**

- Convolutional neural networks (CNNs) and recurrent neural networks (RNNs), two deep learning architectures, dominated machine learning. Their power was demonstrated by innovations in gaming, picture recognition, and natural language processing.

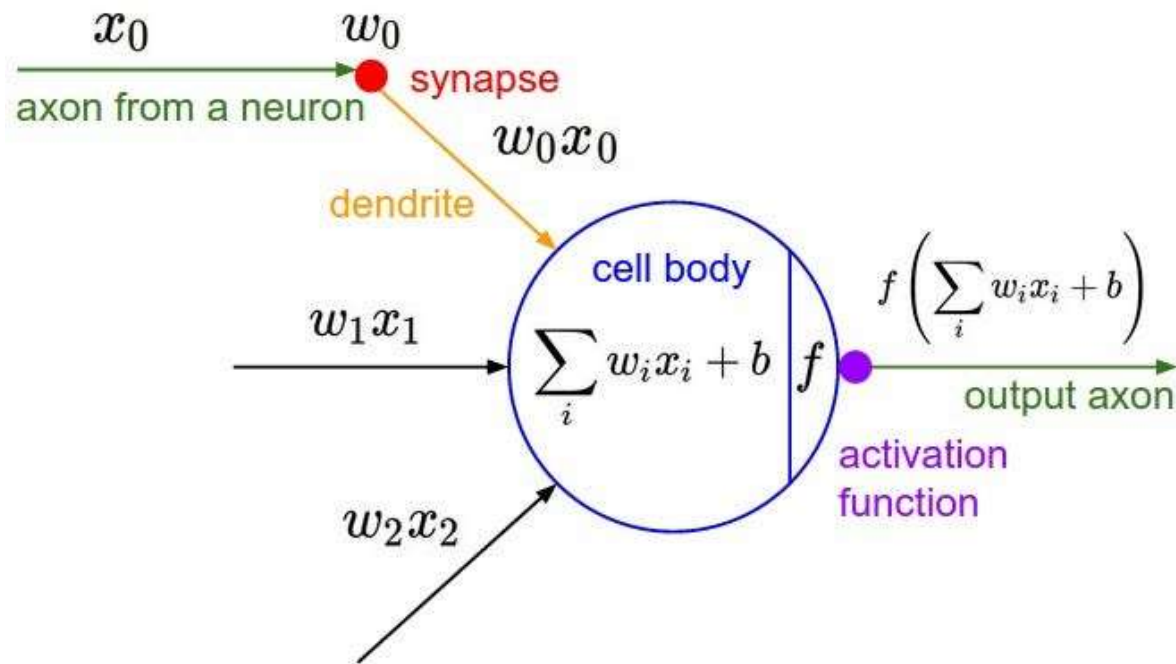
# What are Neural Networks?

- Biological Motivation behind Neural Network is Human Brain.
- Our brain has  $\sim 10^{11}$  neurons, each of which communicates (is connected) to  $\sim 10^4$  other neurons



# Mathematical Model of a Neuron(Perceptron Model)

- Neural networks define functions of the inputs (hidden features), computed by neurons.
- Artificial neurons are called units



# Components of Perceptron Model

## Input Layer:

- The perceptron receives input signals, each associated with a feature of the data. Each input is typically represented as a numerical value.
- **Example:** In an image classification task, the inputs might be pixel values.

## Weights:

- Each input is multiplied by a corresponding weight. Weights determine the importance of each input in the decision-making process.
- **Example:** If a weight is high, the corresponding input has a significant impact on the perceptron's output.

# Components of Perceptron Model

## **Bias:**

- An additional parameter added to the weighted sum of inputs to adjust the output independently of the input values.
- The bias allows the activation function to be shifted left or right, which can be crucial for the model to learn patterns that don't pass through the origin.
- **Example:** If the bias is positive, it can increase the likelihood of the perceptron activating (producing a positive output).

# Components of Perceptron Model

## Summation Function:

- The summation function calculates the weighted sum of the inputs plus the bias. This forms the basis for the next step.

- **Formula:**

$$z = \sum_{i=1}^n w_i x_i + b,$$

- Where,
- $w_i$  are the weights,
- $x_i$  are the inputs,
- and  $b$  is the bias.

# Components of Perceptron Model

## **Activation Function:**

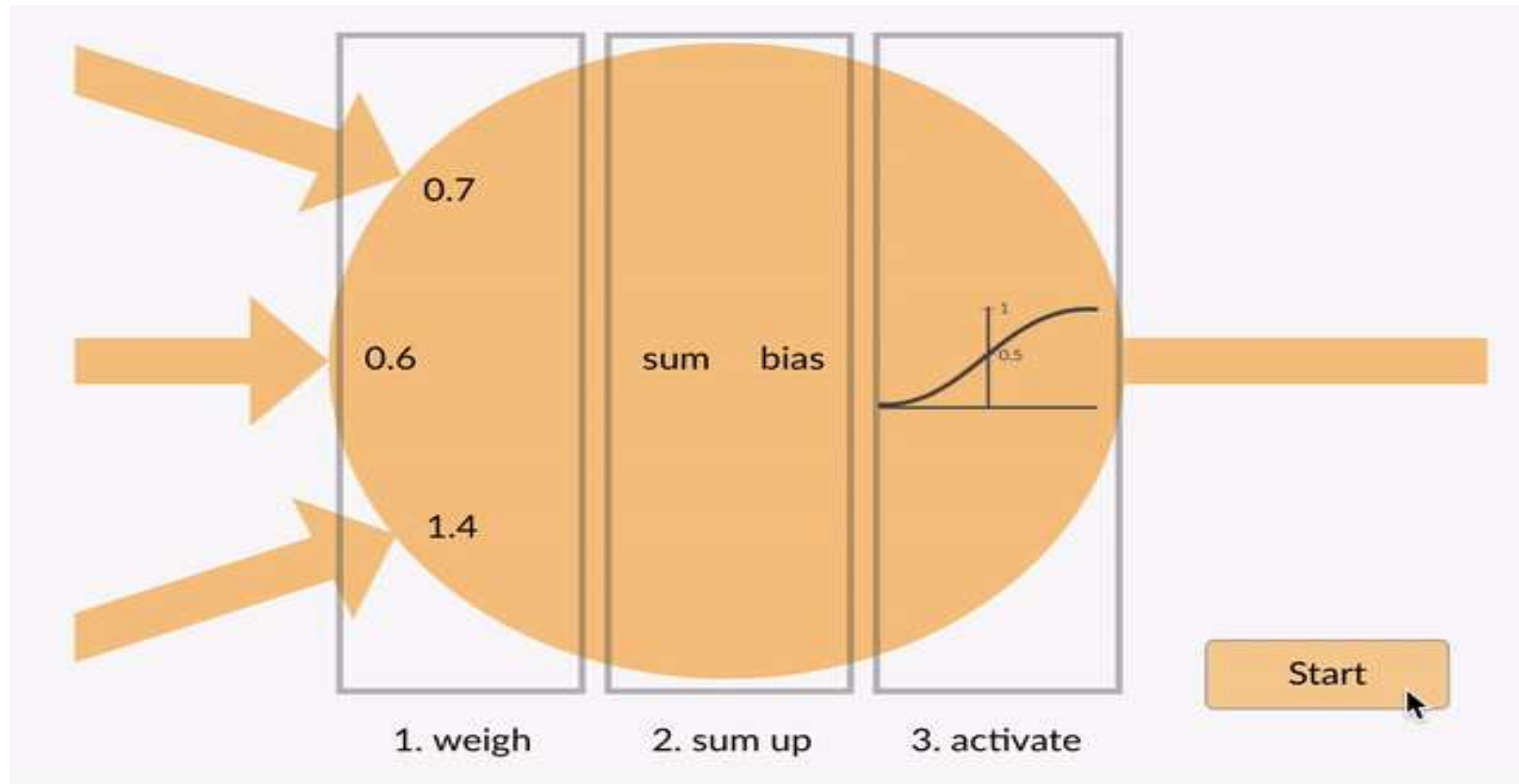
- The activation function takes the summation result and applies a non-linear transformation to decide whether the perceptron should activate (fire) or not. Common activation functions include the step function, sigmoid, ReLU, and tanh.
- **Example:** In a binary classification task, a step function might output 1 if the input exceeds a threshold and 0 otherwise.

# Components of Perceptron Model

## Output:

- The output is the final result after the activation function is applied. In a binary classification problem, the output is typically 0 or 1, representing different classes.
- **Example:** If the task is to determine whether an email is spam or not, the output might be 1 for spam and 0 for not spam.

# Perceptron Model





THANK YOU!!